

HOSPITAL PERSONNEL MANAGEMENT SYSTEM

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Executive Summary

This report presents the design and implementation of a Hospital Personnel Management System developed as part of a Database Systems project. The purpose of the system is to improve the management of hospital staff information, attendance, departments, qualifications, roles, and work shifts using a computerized database system.

Hospitals handle large amounts of personnel data daily. Managing this information manually can result in delays, data redundancy, poor record keeping, and inefficient retrieval of staff information. The Hospital Personnel Management System was therefore developed to provide a centralized and efficient way of storing and managing personnel records.

The system supports operations such as recording staff details, assigning departments and roles, tracking attendance, managing staff qualifications, and generating reports. The database was designed using relational database concepts including entities, attributes, primary keys, foreign keys, and entity relationships.

The project involved requirements analysis, logical database design, physical database implementation using SQL, data population, and query testing. Several tables were created including Personnel, Department, Role, Qualification, Attendance, Shifts, and PersonnelQualifications.

The implemented system demonstrates how database systems can improve organizational efficiency through better data management, reduced redundancy, improved consistency, and faster access to information.

1. Business Case

1.1 Introduction

Healthcare institutions require effective systems for managing personnel information. In many hospitals, staff records are still managed manually using paper files or disconnected systems. This creates challenges such as difficulty in retrieving information, duplication of records, poor attendance tracking, and inefficient management of staff schedules.

The Hospital Personnel Management System was developed to address these challenges by providing a centralized database solution for managing hospital personnel information.

1.2 Application Domain

The Hospital Personnel Management System is designed to support hospital administration and human resource management operations.

The system supports the following tasks:

- Managing hospital personnel records
- Assigning staff to departments
- Managing staff roles and responsibilities
- Recording attendance information
- Managing work shifts
- Managing personnel qualifications
- Generating reports about personnel and attendance

The primary users of the system include:

- Hospital administrators
- Human resource officers
- Department supervisors

- Attendance officers

The system helps users to store, retrieve, update, and manage personnel information efficiently.

1.3 Analysis of Benefits

Tangible Benefits

The system provides several measurable benefits including:

- Faster retrieval of personnel records
- Reduced paperwork and manual filing
- Improved attendance tracking
- Reduced data redundancy
- Better organization of personnel information
- Easier generation of staff reports
- Improved data consistency and accuracy

Intangible Benefits

The system also provides non-measurable benefits such as:

- Improved decision making
 - Better staff management
 - Improved organizational image
 - Increased efficiency among administrators
 - Enhanced accountability of personnel
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1.4 Analysis of Costs

Tangible Costs

The implementation of the system may require:

- Computers and hardware
- Database software
- Electricity and internet connectivity
- System maintenance costs

- Training costs for users

Intangible Costs

Other costs may include:

- Time required to train users
- Resistance to adopting new technology
- Temporary reduction in productivity during implementation

Despite these costs, the long-term benefits outweigh the expenses, making the system economically feasible.

1.5 User Issues

Several user-related issues may arise during implementation of the system.

Skills Required

Users should possess basic computer literacy skills including:

- Operating a computer
- Navigating software interfaces
- Entering and retrieving data

Adoption Effort

Users may initially face challenges adapting from manual systems to computerized systems. Training and user support will therefore be necessary.

Security Awareness

Users should understand the importance of securing login credentials and protecting sensitive personnel information.

1.6 Risk Assessment

Technical Risks

- Hardware failure
- Database corruption

- Power outages
- Software bugs
- Loss of data

Organizational Risks

- Resistance to change
- Poor user adoption
- Inadequate training
- Unauthorized access to records

Mitigation Measures

The following measures can help reduce risks:

- Regular database backups
 - User training
 - Access control mechanisms
 - Antivirus and security software
 - Proper documentation
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1.7 Limitations of the System

Although the system improves personnel management, it has several limitations:

- The system does not handle payroll processing
 - The system does not manage patient information
 - Internet or electricity interruptions may affect usage
 - The system requires trained users
 - The system may require future upgrades for additional functionality
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2. Design

2.1 Logical Design

2.1.1 List of Entities

The system contains the following entities:

1. Personnel
 2. Department
 3. Role
 4. Shifts
 5. Qualification
 6. Attendance
 7. PersonnelQualifications
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2.1.2 Entities and Attributes

Personnel

Attribute	Description
PersonnelID	Unique identifier for personnel
FirstName	Personnel first name
LastName	Personnel last name
Gender	Personnel gender
Phone	Personnel phone number
Email	Personnel email address
HireDate	Date personnel was hired
DepartmentID	Department assigned
RoleID	Role assigned
ShiftID	Shift assigned

Department

Attribute	Description
DepartmentID	Unique department identifier
DepartmentName	Name of department

Role

Attribute	Description
RoleID	Unique role identifier
RoleName	Name of role

Shifts

Attribute	Description
ShiftID	Unique shift identifier
ShiftName	Name of shift
StartTime	Shift start time
EndTime	Shift end time

Qualification

Attribute	Description
QualificationID	Unique qualification identifier
QualificationName	Qualification name
Institution	Institution attended
YearObtained	Year qualification obtained

Attendance

Attribute	Description
AttendanceID	Unique attendance identifier
DateRecorded	Attendance date
ToBeFilled	Attendance status
PersonnelID	Personnel reference

PersonnelQualifications

Attribute	Description
QualificationID	Qualification reference
PersonnelID	Personnel reference

2.1.3 Entity Relationships

The system contains the following relationships:

Department and Personnel

One department can have many personnel.

Relationship Type: One-to-Many (1:M)

Role and Personnel

One role can be assigned to many personnel.

Relationship Type: One-to-Many (1:M)

Shifts and Personnel

One shift can be assigned to many personnel.

Relationship Type: One-to-Many (1:M)

Personnel and Attendance

One personnel member can have many attendance records.

Relationship Type: One-to-Many (1:M)

Personnel and Qualification

One personnel member can have multiple qualifications, and one qualification can belong to multiple personnel.

Relationship Type: Many-to-Many (M:M)

This relationship is resolved using the PersonnelQualifications table.

2.1.4 Multiplicities

The multiplicities in the system are:

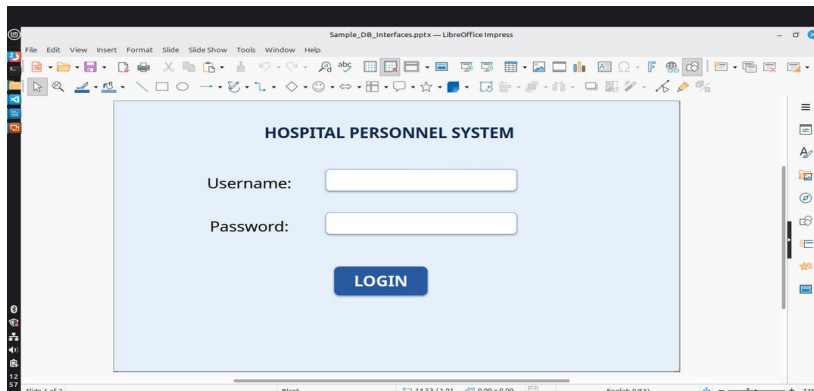
Relationship	Multiplicity
Department to Personnel	1:M
Role to Personnel	1:M
Shifts to Personnel	1:M
Personnel to Attendance	1:M
Personnel to Qualification	M:M

2.1.5 User Interface Design

The system provides interfaces for managing personnel records, attendance, qualifications, departments, and reports.

The following interface representations were designed:

- Login Interface
- Dashboard Interface
- Add Personnel Interface
- Attendance Interface
- Reports Interface



Dialogue Structure

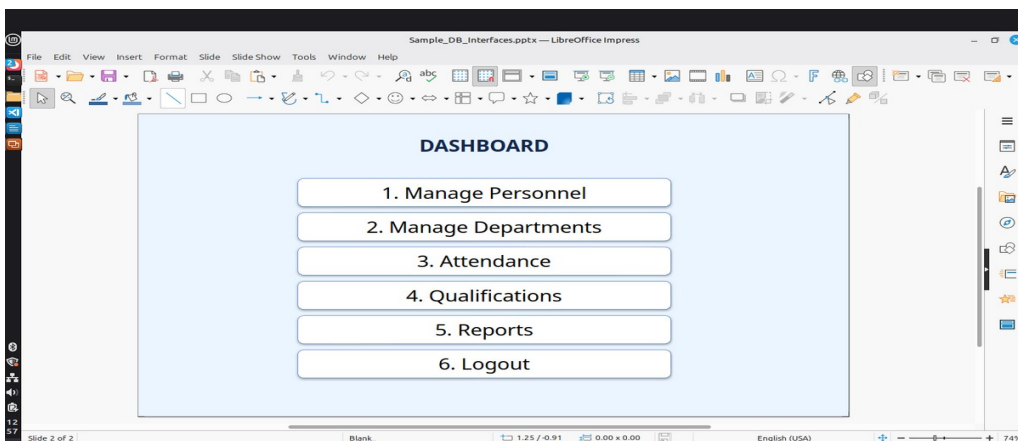
The dialogue structure of the system is shown below:

Login



Dashboard

- ├─ Manage Personnel
 - ├─ Add Personnel
 - ├─ Update Personnel
 - └─ Delete Personnel
- ├─ Attendance
 - ├─ Record Attendance
 - └─ View Attendance
- ├─ Qualifications
 - ├─ Add Qualification
 - └─ View Qualifications
- ├─ Reports
 - ├─ Personnel Reports
 - └─ Attendance Reports



2.2 Physical Design

2.2.1 Database Schema

The physical database design was implemented using SQL tables with primary keys and foreign keys.

The database contains the following tables:

- Department
- Role
- Shifts
- Qualification
- Personnel
- Attendance
- PersonnelQualifications

The relationships among the tables were implemented using foreign key con